EXPERIMENT 1

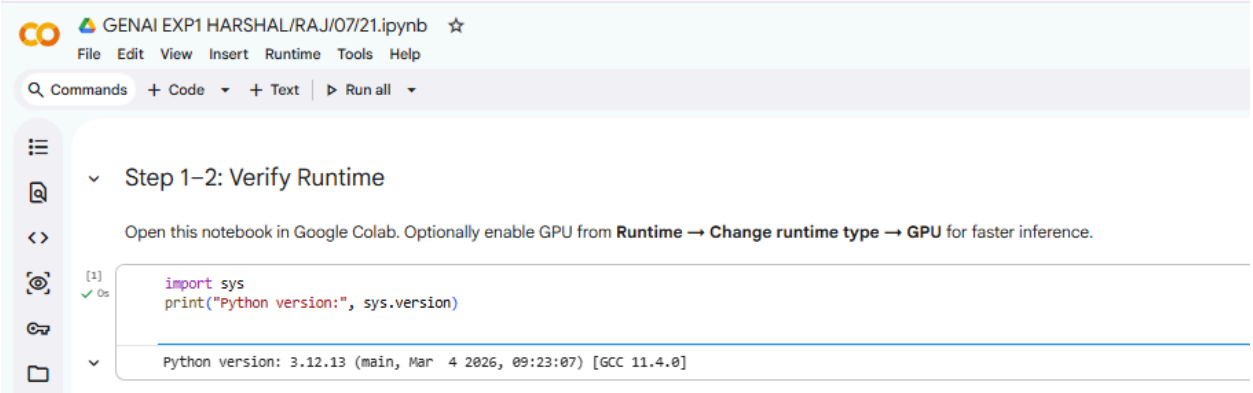
AIM : Getting Started with Python, TensorFlow and Hugging Face Transformers

THEORY : Generative AI refers to artificial intelligence systems capable of creating new content such as text, images, audio, and code by learning patterns from large datasets.

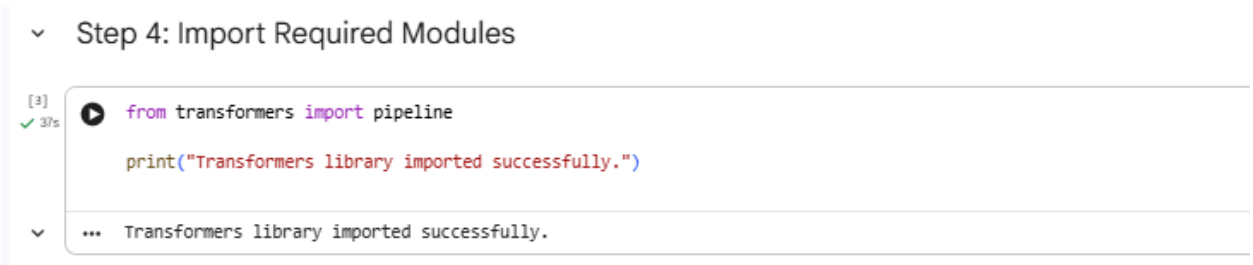
- Python is the most widely used programming language for AI due to its simplicity and extensive ecosystem of libraries.
- Google Colab provides a cloud-based Jupyter Notebook environment with free GPU support, making it ideal for AI experimentation without local installation.
- TensorFlow and PyTorch are popular deep learning frameworks used to build and deploy neural networks. Hugging Face Transformers provides access to thousands of pre-trained transformer models.

COLAB FILE : GENAI EXP1 HARSHAL/RAJ/07/21.ipynb

CODE AND OUTPUT :







Step 5: Create a Text-Generation Pipeline

[4] ✓ 12s

```
# Load GPT-2 text generation model
generator = pipeline("text-generation", model="gpt2")
print("GPT-2 text-generation pipeline created successfully.")
```

```
config.json: 100% 665/665 [00:00<00:00, 33.0kB/s]
Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to enable higher rate limits and faster downloads.
WARNING:huggingface_hub.utils._http:Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to enable higher rate limits and faster downloads.
model.safetensors: reconstructing file: 100% 548MB / 548MB, 48.3MB/s
model.safetensors: downloading bytes: 474MB, 40.9MB/s
Loading weights: 100% 148/148 [00:00<00:00, 2943.62it/s]
generation_config.json: 100% 124/124 [00:00<00:00, 10.2kB/s]
tokenizer_config.json: 100% 26.0/26.0 [00:00<00:00, 2.08kB/s]
vocab.json: 100% 1.04M/1.04M [00:00<00:00, 6.82MB/s]
merges.txt: 100% 456k/456k [00:00<00:00, 2.42MB/s]
tokenizer.json: 100% 1.36M/1.36M [00:00<00:00, 5.97MB/s]
GPT-2 text-generation pipeline created successfully.
```

Step 6: Execute a Sample Generation

[5] ✓ 26s

```
output = generator(
    "Generative AI is transforming education because",
    max_length=60,
    num_return_sequences=1
)
print(output[0]["generated_text"])
```

```
[transformers] Passing `generation_config` together with generation-related arguments=({'max_length', 'num_return_sequences'}) is deprecated and will be removed in future versions. Please pass either a `generati
[transformers] Both `max_new_tokens` (=256) and `max_length`(=60) seem to have been set. `max_new_tokens` will take precedence. Please refer to the documentation for more information. (https://huggingface.co/doc
[transformers] Ignoring clean_up_tokenization_spaces=True for BPE tokenizer GPT2Tokenizer. The clean_up_tokenization post-processing step is designed for WordPiece tokenizers and is destructive for BPE (it strip
Generative AI is transforming education because of its ability to integrate technologies into a larger, more complex system of learning and cognition.

As a result, many of the more innovative ways that human technology has been used to transform education have found their way into the human-machine relationship. For example, when computer programs and training

In some ways, the future of education is an increasingly complex subject, which is what brought together researchers across the globe from the United States and the United Kingdom to develop a new form of AI cal

The new method is called "supervised learning," and it's the most sophisticated in the industry. The researchers created a new, artificial intelligence class called "supervised learning" to test the ability of s

In the first test, the researchers used a new type of computer program that uses a combination of neural network and artificial intelligence to learn the way a human hand works. The program, called "supervised l
```

Step 7: Modify the Prompt — Repeat with Five Different Prompts

[6] ✓ 1m

```
prompts = [
    "Artificial Intelligence will shape the future of",
    "The most important skill for a data scientist is",
    "In the next ten years, robots will",
    "Climate change can be addressed by",
    "The best way to learn programming is"
]

results = {}

for p in prompts:
    output = generator(
        p,
        max_length=60,
        num_return_sequences=1,
        do_sample=True,
        top_k=50
    )
    generated = output[0]["generated_text"]
    results[p] = generated
    print("Prompt:", p)
    print("Generated:", generated)
    print("-" * 80)
```

```
[transformers] Passing `generation_config` together with generation-related arguments={{'do_sample', 'num_return_sequences', 'top_k', 'max_length'}) is deprecated and will be removed in future versio
[transformers] Both `max_new_tokens` (=256) and `max_length`(=60) seem to have been set. `max_new_tokens` will take precedence. Please refer to the documentation for more information. (https://huggin
[transformers] Both `max_new_tokens` (=256) and `max_length`(=60) seem to have been set. `max_new_tokens` will take precedence. Please refer to the documentation for more information. (https://huggin
Prompt: Artificial Intelligence will shape the future of
Generated: Artificial Intelligence will shape the future of social and political life in the United States, even if there is no evidence of it.

We need technology that can predict what humans think of us. Without it, we would never know what we are.

The new research focuses on the use of artificial intelligence to predict people. If the results are correct, it could help people make better decisions about the way they want to live.

They say that we are likely to see what we call "psychological" robots, with human personalities and thoughts, that will make us smarter than our real counterparts.

We might not see what we would like to see, but as technology advances, more and more people will have the ability to understand what we have to offer.

The next step is to make it possible for more people to understand what they think.

This is a very real and potentially very dangerous moment.

It is also a time when people are coming to the realization that this is not just a good thing, this is possible.

"The future of social and political life in the United States is going to be much better if people are able to understand what we are doing," said Dr. Robert M. Zwolinski, a professor of psychology at
--------------------------------------------------------------------------------
[transformers] Both `max_new_tokens` (=256) and `max_length`(=60) seem to have been set. `max_new_tokens` will take precedence. Please refer to the documentation for more information. (https://huggin
Prompt: The most important skill for a data scientist is
Generated: The most important skill for a data scientist is understanding the data and the process of analyzing it. There are several ways to do this that is unique for a data scientist, but I think i

The most important skill for a data scientist is understanding the data and the process of analyzing it. There are several ways to do this that is unique for a data scientist, but I think it is the mo

If you have an understanding of data, you can use that knowledge to interpret it and interpret it in a very specific way. In general, you need to understand how data is structured. And then you need t
```

[7] ✓ 1m

```python
prompt = "Artificial Intelligence will shape the future of"

param_sets = [
    {"max_length": 40,  "temperature": 0.3, "do_sample": True, "top_k": 50},
    {"max_length": 60,  "temperature": 0.8, "do_sample": True, "top_k": 50},
    {"max_length": 80,  "temperature": 1.2, "do_sample": True, "top_k": 50},
    {"max_length": 60,  "temperature": 0.8, "do_sample": True, "top_k": 10},
    {"max_length": 60,  "temperature": 0.8, "do_sample": False, "top_k": 50},
]

for i, params in enumerate(param_sets, 1):
    result = generator(prompt, num_return_sequences=1, **params)
    print(f"Configuration {i}: {params}")
    print("Generated:", result[0]["generated_text"])
    print("-" * 80)
```

```
So, we are here to discuss in detail this new way of thinking about Artificial Intelligence. We are here to discuss the big picture. We are here to discuss the great possibilities in it
--------------------------------------------------------------------------------
[transformers] Both `max_new_tokens` (=256) and `max_length`(=60) seem to have been set. `max_new_tokens` will take precedence. Please refer to the documentation for more information. (
Configuration 3: {'max_length': 80, 'temperature': 1.2, 'do_sample': True, 'top_k': 50}
Generated: Artificial Intelligence will shape the future of our lives through a network that interacts with robots or robots interact with us through artificial intelligence technology.

In many ways, humanity's evolution requires us to live in a virtual environment, which, in turn, introduces significant challenges to AI. For instance, artificial learning software on s

If machines, including machines that run "machine learning," are to be built into everything being played today, technology that makes a user of an app able to see any individual's phys
--------------------------------------------------------------------------------
[transformers] The following generation flags are not valid and may be ignored: ['temperature']. Set `TRANSFORMERS_VERBOSITY=info` for more details.
[transformers] Both `max_new_tokens` (=256) and `max_length`(=60) seem to have been set. `max_new_tokens` will take precedence. Please refer to the documentation for more information. (
Configuration 4: {'max_length': 60, 'temperature': 0.8, 'do_sample': True, 'top_k': 10}
Generated: Artificial Intelligence will shape the future of the human race.
```

```
The first step will be to develop AI technologies that would help humans learn from other people and understand the world around them. The second step is a new kind of AI that

We will be able to use AI to help us understand the world and understand ourselves.

AI will help to make us more like our real selves in the future.

We will be able to build more artificial intelligence and machines to help us solve problems.

The AI revolution is being developed in the United States and China.

In the next 10 years, more and more people will be able to become part of the human race and become more like the real person in the world.

The AI revolution is being developed in the United States and China.In the next 10 years, more and more people will be able to become part of the human race and become more li

We will be able to use AI to help us understand
--------------------------------------------------------------------------------
Configuration 5: {'max_length': 60, 'temperature': 0.8, 'do_sample': False, 'top_k': 50}
Generated: Artificial Intelligence will shape the future of the world.

The future of the world is not yet clear. But it is clear that AI will be the next big thing.

The future of the world is not yet clear. But it is clear that AI will be the next big thing.

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```

Step 9: Record Outputs and Compare Quality of Responses

```
[8]
✓ 0s
import pandas as pd

comparison_df = pd.DataFrame({
    "Prompt": list(results.keys()),
    "Generated Text": list(results.values())
})

comparison_df
```

	Prompt	Generated Text
0	Artificial Intelligence will shape the future of	Artificial Intelligence will shape the future ...
1	The most important skill for a data scientist is	The most important skill for a data scientist ...
2	In the next ten years, robots will	In the next ten years, robots will be driving ...
3	Climate change can be addressed by	Climate change can be addressed by a combinati...
4	The best way to learn programming is	The best way to learn programming is to learn ...

OBSERVATIONS :

- **Lower temperature values** (e.g., 0.3) produce more conservative, repetitive, and predictable text, staying close to the most probable next words.
- **Higher temperature values** (e.g., 1.2) produce more diverse and creative — but sometimes less coherent — text, as the model samples from a wider probability distribution.
- **do_sample=False** (greedy decoding) always produces the same deterministic output for a given prompt, whereas do_sample=True introduces randomness and variety across runs.
- **A smaller top_k** (e.g., 10) restricts the model to choosing from fewer candidate words at each step, producing more focused text, while a larger top_k allows more variety.
- **max_length** controls how much text is generated; longer generations sometimes drift off-topic since GPT-2 has no long-range planning ability.
- Overall, GPT-2 produces fluent but not always factually accurate text, illustrating both the capability and the limitations of early transformer-based generative language models.

CONCLUSION : This experiment demonstrated how to set up a Generative AI development environment in Google Colab using Python, TensorFlow, PyTorch, and the Hugging Face Transformers library. A pre-trained GPT-2 model was used to generate text from multiple prompts, and the effect of parameters such as max_length, temperature, do_sample, and top_k on the quality and diversity of generated text was observed and analyzed. This forms the foundation for working with larger and more advanced transformer-based Large Language Models (LLMs).